

Lloyd Ntim

Senior Full Stack Engineer | AI & Product Development

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PROFILE

Senior Full-Stack Engineer with 15 years of experience building and modernising customer-facing software across enterprise and startup environments.

My core expertise is in React/TypeScript application architecture and full-stack development with Node.js, Python and relational data systems, using spec-driven development and agentic workflows (Claude Code) to take products from ambiguous requirements through technical design, implementation and delivery.

I focus on creating maintainable foundations that help teams ship reliably and evolve products across the full stack.

SELECTED ACHIEVEMENTS

Language-learning app delivery: At AnzaKen, led the redesign of an existing vocabulary-learning prototype into a live full-stack product, owning system architecture, data modelling, API design, frontend and UX. Validated the core journey with ~20 testers and established the foundation for subscriptions and AI-enabled learning features, using spec-driven and agentic workflows to accelerate delivery.

Multi-market frontend architecture: At Vorwerk, designed and established a new React-based frontend architecture for the AEM CMS platform, resolving a key architectural challenge around improving the website's Lighthouse performance scores from 75 to ~85-90+ and preserving SEO capabilities while maintaining WYSIWYG authoring for content editors and creating the foundation for the wider vorwerk.com platform.

End-to-end product engineering: At Guilds Inc, helped take a crypto gaming wallet from early development to beta release, delivering the product for both mobile and web while contributing across frontend and backend. Translated the CTO's technical vision and the Art Director's designs into a working application and collaborated closely with the COO on product direction through to launch.

Shared component library: At E.ON, helped establish a reusable component library used across multiple AEM-based country websites, replacing fragmented legacy implementations with a centrally maintained system. Later led the delivery of a separate component library for E.ON.com into production.

Advisor platform: At Deutsche Bank, joined as the project's second engineer and helped take an early proof of concept into a live wealth-management platform used by ~50–100 internal advisors, establishing the React/Redux architecture that supported the product as the engineering team grew from 3 to ~25 and was later reused by a second department for a related platform.

CORE STRENGTHS

PRODUCT ENGINEERING & DELIVERY: End-to-end product development from early definition through production | Balancing UX, technical quality and delivery constraints | Building and evolving products across startup and enterprise environments

FULL-STACK ARCHITECTURE & SYSTEM DESIGN: Frontend, backend and data architecture | API and service-layer design | Reusable components and shared application foundations | Performance, SSR and Headless CMS | Systems built for reuse across teams and markets

AGENTIC ENGINEERING & WORKFLOWS: Reusable AI Skills | Context engineering | LLM evaluation & regression testing | Agent & tool orchestration | Model failure modes & guardrails | Claude Code, Codex & Gemini | Spec-driven development

TECHNICAL LEADERSHIP & STAKEHOLDER ALIGNMENT: Working with clients, product owners, designers and architects | Translating product needs into technical direction | Mentoring, workshops, documentation and knowledge sharing

PROFESSIONAL EXPERIENCE

ANZAKEN

London, remote | Aug 2025 – present

Senior Product Engineer (Contract)

Startup business consultancy delivering digital products and technical solutions for clients. Working across different product initiatives, with current focus on a vocabulary-learning application taken from prototype through architecture, implementation, user validation and production delivery, alongside internal tooling and client-facing technical consulting.

- **Product delivery:** Built and launched a vocabulary app combining translation, text-to-speech, flashcard practice and gamified exercises, allowing users to study, practise and retain new words and phrases while tracking their progress over time.
- **Application architecture:** Modernised the app's original React (JavaScript)/MongoDB-based version with TypeScript, Next.js SSR, Express and PostgreSQL, improving maintainability, performance and data modelling while introducing layered backend architecture and OWASP-aligned security practices.
- **Engineering workflow:** Apply agentic development workflows across implementation, testing and code review, orchestrating Claude, OpenAI, Gemini and local models via OpenRouter and Ollama and integrating custom MCP services; introduced reusable, evaluated Skills and shared workflow patterns to improve consistency across development tasks.
- **User validation:** Ran structured testing with around 20 students, interpreters and colleagues across the complete learning journey, using observed behaviour and feedback to identify usability issues and guide product improvements.
- **Client discovery:** Participated in client pitches and discovery discussions, translating business needs into technical requirements, assessing feasibility and delivery scope, and helping shape practical solution proposals for prospective clients.

Stack: React, TypeScript, Next.js, Node.js, Express, REST APIs, PostgreSQL, GitHub Actions, Claude Code, Codex.

VORWERK

Zürich, remote | Jul 2024 – Jun 2025

Frontend Architect (Contract)

German consumer-products company operating international brands including Thermomix and Kobold. Designed the frontend architecture for vorwerk.com across four European markets, solving the challenge of improving performance and SEO while preserving the AEM authoring workflows used by content teams.

- **Frontend architecture:** Evaluated multiple frontend frameworks and architectural approaches, including Next.js and Web Components, against performance, maintainability and development complexity. Documented the outcome in an Architecture Decision Record (ADR), establishing React with Vite as the basis for the new frontend architecture.
- **Backend integration:** Worked across the backend and frontend, analysing Java models and servlets, defining JSON data flows and integrating CMS-backed services with server-side rendering. Added JUnit tests for backend logic and coordinated data contracts and rendering behaviour to keep server and client output consistent.
- **Product & UX refinement:** Worked with the Product Owner, Business Analyst and design/art team to shape user stories, assess technical feasibility and refine usability and accessibility requirements, including WCAG 2.2-aligned improvements.
- **Serverless deployment:** Worked closely with Solution and Technical Architects to implement React/Vite server-side rendering within an AWS Lambda-based serverless environment, integrating with AEM to preserve SEO-critical content and improve initial page-load performance.
- **Team enablement:** Created the boilerplate and reference codebase for the new frontend setup, wrote implementation and environment setup guides, and presented the approach during a three-day workshop that enabled up to 20 developers to start building components, features and pages with the new solution.

Stack: React, TypeScript, Node.js, Java, AEM, Vite SSR, JSON, AWS Lambda, GitLab CI/CD, Lighthouse.

AOE – CLIENT: KIA

Wiesbaden, remote | Oct 2023 – Mar 2024

Senior Web Developer (Contract)

German digital consultancy delivering custom software and digital products. End client: Kia, a global automotive brand. Senior engineer on Kia's public website and dealership-advisor workflows, contributing across frontend/backend integration and mentoring junior developers.

- **Customer journey improvement:** Resolved a long-standing routing issue in the advisor and store-locator flow, aligned the experience with Kia's design requirements and delivered the approved customer journey.
- **Full-stack integration:** Built Node.js BFF logic to transform dealer and appointment data for React features and updated Java-based AEM models to expose consistent frontend data across application layers.
- **Testing & code quality:** Introduced Jest, linting, structured reviews and component standards, increasing frontend test coverage from near zero to ~70% and making defects easier to isolate and resolve.
- **Mentoring & documentation:** Created technical guidance and mentored junior developers on React best practices, test-driven development and component design.

Stack: React, TypeScript, Node.js, Express.js, BFF, Java, AEM, REST APIs, Maven, Jest, Figma, Git.

NATIVE INSTRUMENTS

Berlin, remote | Nov 2022 – Mar 2023

Full-Stack Engineer (Contract)

Music technology company. Brought in to deliver SEO-focused improvements across the main product website, working across the full stack with frontend and backend engineers and SEO specialists in Kubernetes- and Helm-managed environments.

- **Pagination SEO enhancement:** Improved React Query and GraphQL pagination so product pages were server-rendered and URL-addressable, allowing search crawlers to index catalogue pages while preserving the existing scroll-based experience.
- **Product catalogue rendering fixes:** Introduced generic typing for a React/TypeScript and GraphQL data accessor, exposing and correcting a null-handling defect across five production rendering paths behind broken product links and malformed content.
- **Mentoring & code quality:** Supported a junior developer through code reviews, debugging and feature implementation, helping to maintain consistency and engineering quality across teams.
- **Release operations:** Worked with Google Cloud-hosted Kubernetes environments, using Helm to inspect release history, troubleshoot deployments and perform direct rollbacks when required.

Stack: React, TypeScript, Node.js, React Query, GraphQL, Apollo, Google Cloud, Kubernetes, Helm, Git.

GUILDS INC

London, remote | Mar 2022 – Oct 2022

Product Engineer (Contract)

Seed-stage gaming technology startup. Brought on as a product engineer within a small founding team to help take a mobile and web crypto gaming wallet from early development through testing and release, working closely with CTO, COO and art director to translate product requirements into features.

- **Cross-platform app delivery:** Built the React Native iOS and web wallet across ~20 screens, including onboarding, navigation and the dashboard, using shared Styled Components, Recoil state management and WebSockets for real-time client updates.
- **Game prototype:** Built an interactive Swift/SwiftUI game prototype to validate wallet functionality and test the end-to-end rewards experience before release.
- **Backend & data delivery:** Developed a FastAPI transaction endpoint, updated the related PostgreSQL data model and integrated the flow into the frontend; also enhanced the phone-verification API alongside UX improvements.
- **Third-party integration:** Integrated Intercom messaging and notifications into the app, working directly with the provider's engineering team to diagnose and resolve integration issues.

Stack: React Native, TypeScript, Swift, SwiftUI, Python, FastAPI, PostgreSQL, SQL, pgAdmin, Git.

E.ON

Essen | Nov 2018 – Feb 2022

European energy company. Initially joined as a Frontend Developer to migrate E.ON's legacy collection of CommonJS-based AEM components to a more modern React and Web Components approach, standardising multiple versions of similar components; later returned as Lead Developer with broader ownership of the component library used across multiple markets.

Lead Developer (Contract) | Nov 2020 – Feb 2022

- **Multi-market component library:** Maintained and extended E.ON's wider 60-component library across Germany, Hungary, the Czech Republic, Slovakia, Romania and Italy, consolidating duplicated legacy implementations into a centrally maintained system.
- **E.ON.com component library delivery:** Led the implementation of a separate component library for the official E.ON.com website, with technical responsibility across the full component collection (approx. 15–20) while also building and reviewing components.
- **Team leadership:** Guided two developers through implementation and reviews while coordinating work across frontend, backend and AEM.

Frontend Developer (Contract) | Nov 2018 – Mar 2020

- **Component modernisation:** Led the conversion of ~30 jQuery-based, and partially duplicated, legacy AEM components into reusable React components; helped to review and assess the first few Stencil.js-based Web Components to be part of a multi-market component library.
- **Data and component integration:** Developed Java models and XML authoring dialogs to connect AEM-managed content with React components while preserving the existing functionality used by content editors.
- **Build & debugging:** Used Maven-based AEM workflows, supported environment deployments and diagnosed issues across Java-based servlets, CMS content and reusable frontend components.

Stack: AEM, Java, Maven, React, Node.js, Stencil.js, TypeScript, Web Components, Jest, Cypress, Docker, Git.

HEYCAR

Berlin, remote | Jun 2020 – Oct 2020

Full-Stack Engineer (Contract)

Automotive marketplace. Joined the cross-functional team delivering HeyCar's Spanish-market launch, contributing across customer journeys, backend integration, localisation and testing.

- **Market launch:** Shipped the core vehicle-search, listing and dealer journeys through production by the committed launch deadline.
- **Node.js BFF:** Built models and transformation logic to adapt raw vehicle records from upstream Kotlin APIs for vehicle detail pages in the frontend.
- **CMS implementation & testing:** Designed Contentful models and dynamic routing so editors could manage Spanish-market content, while covering APIs, search, routing, listings and React components with integration tests.

Stack: React, Next.js, Node.js, Express.js, HTML5/CSS3, Contentful CMS, Cypress, Figma, Kotlin, Docker, Git.

LEFTSHIFT IT

London, remote | May 2018 – Oct 2018

Web Developer (Contract)

Fintech consultancy. Owned UX and frontend delivery for a financial-data application rebuild, working closely with the Solution Architect from interface design through frontend architecture and implementation.

- **UX & user journeys:** Defined user journeys, interaction flows and interface concepts with the Solution Architect, turning business requirements into clearer application workflows.
- **Frontend engineering:** Built the application frontend around complex financial data, creating reusable components, integrating backend APIs and structuring data-heavy interfaces for maintainability and reuse.
- **Technical evaluation:** Developed proofs of concept with Vue, React and Knockout.js to evaluate frontend approaches to make architectural decisions with stakeholders.

Stack: HTML5/CSS3, ES6, Vue, React, Knockout, WebPack, Balsamiq, Sketch, Git.

DARK BLUE

London, remote | Jul 2017 – Mar 2018

Full-Stack Developer (Contract)

Creative agency. End client: Harness Properties, a London property agency. Delivered the React-based frontend for the client website, working with designers and client stakeholders from requirements through implementation.

- **React migration:** Reworked the existing frontend into a React/Redux-based application, replacing legacy styling and implementation patterns.
- **API Improvements:** Added rate limiting, request timeouts, centralised error handling and API-level integration tests to BFF endpoints related to listing, search and location data.
- **Cross-team delivery:** Worked with project managers, business stakeholders and backend engineers to align frontend implementation with product requirements.

Stack: React, Redux, Next.js, Node.js, Redis, JavaScript/ES6, Jasmine, HTML5/CSS3, Webpack, Git.

DEUTSCHE BANK

Frankfurt, remote | Feb 2016 – Feb 2017

Senior Frontend Developer (Contract)

Global banking and financial services company. Joined a three-person team to take an internal wealth-management application used by ~50–100 internal advisors from proof of concept to production, contributing across the full stack and data.

- **Application foundation:** Helped establish the React and Node.js-based application architecture; implemented routing, authentication, Redux state management, testing and error handling.
- **Backend & data:** Developed and integrated Node.js-based REST APIs and PostgreSQL-backed data models, creating test data for client details and related wealth-management use cases.
- **Architecture reuse:** Produced architecture and implementation documentation that enabled a second Deutsche Bank team to build a related application from the same foundation.
- **Team scaling:** Contributed to growing the team from 3 to ~25 engineers by creating technical assessments, interviewing candidates, training developers and leading React workshops.

Stack: React, Redux, Mocha, Node.js, PostgreSQL, Less, Webpack, Git.

EARLIER ROLES

- **Web Developer** — Waitrose (Bracknell, 2015–2016) · Backbone, Node.js
- **Web Developer** — AKQA (London, 2014–2015) · AEM/CQ5, JavaScript
- **Frontend Developer** — Ryanair (Dublin, 2014) · Backbone, Angular, Sass
- **UI Developer** — Skyscanner B2B (Edinburgh, 2014) · JavaScript, jQuery, ASP.NET
- **JavaScript Developer** — Sony Europe (London, 2013) · TypeScript, Facebook API
- **Earlier** — First Clarity, Smile-on (London, 2012–2013) · JavaScript, Backbone, PHP

TECHNOLOGY STACK

- **Languages:** TypeScript, JavaScript, Python, Java, SQL
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST APIs, GraphQL, OpenAPI, JWT
- **Databases & Data:** PostgreSQL, MongoDB, Redis, Prisma, Mongoose
- **Frontend:** React, Next.js, React Native, Swift/SwiftUI, HTML5, CSS3
- **Testing:** Vitest, Jest, Playwright, Cypress, React Testing Library
- **Cloud & DevOps:** AWS, GCP, Kubernetes, Helm, Docker, GitHub Actions, GitLab CI/CD, Vercel
- **Build & Tooling:** Vite, Webpack, Maven, pnpm
- **AI Development:** Claude Code, Codex, Copilot, agentic workflows, spec-driven development
- **CMS & Platforms:** AEM, Contentful
- **Observability & Analytics:** Sentry, PostHog, Google Analytics
- **UI & Quality:** Tailwind CSS, Styled Components, Storybook, WCAG, Lighthouse

SELECTED PROJECTS

AI Skills Framework | Agentic AI, Context Engineering & LLM Evaluation 2026–present
github.com/lloydntim/ai-skills-framework

Built a reusable TypeScript framework for developing, testing and evaluating AI Skills as engineered workflows rather than standalone prompts. Implements deterministic validation, generate/validate/revise pipelines, benchmark and regression evaluation, A/B and pairwise comparison, prompt/dataset versioning, context and token/cost instrumentation, and privacy controls separating reusable public Skills from private data. Presented the framework internally and enabled colleagues to use and collaborate on shared Skills.

Crypto Asset Tracker | Crypto portfolio tracker 2023–present
github.com/lloydntim/crypto-asset-tracker-client

Full-stack portfolio product with a React/TypeScript UI and Node.js/Express backend, Apollo GraphQL data layer, MongoDB/Mongoose persistence, JWT/bcrypt authentication and CoinMarketCap integration. Built multi-wallet valuation logic, reusable atomic-design components, unit/E2E coverage, logging and containerised/deployed services across Vercel and Render.

EDUCATION

BA (Hons) Digital Media and Marketing | London Metropolitan University 2005 – 2008
Abitur | Bürgermeister Smidt Gymnasium, Bremerhaven 2001 – 2004

LANGUAGES AND AVAILABILITY

Languages | English and German bilingual; French and Spanish fluent

Availability | Available for permanent and contract roles across DACH and remote